

Laboratory Guide 3.5: Grover's Algorithm

Quantum Search and Amplitude Amplification

Course: Quantum Computing Foundation

1 Introduction and Objectives

In Lab 3, we used an Oracle to discover a global property of a function (Constant vs. Balanced). In this laboratory, we tackle a much more practical problem: **Unstructured Search**. Imagine looking for a specific name in a phone book that is completely scrambled. We will build **Grover's Algorithm**, which uses quantum interference to find the needle in the haystack quadratically faster than any classical computer.

Prerequisites:

- Lesson QC14: *Quantum Algorithms (Grover & Shor)*
- Lab 3: *Oracles and Phase Kickback*

Learning Objectives:

1. Understand the two components of Grover's Algorithm: The Oracle and the Diffuser.
2. Mathematically track "Inversion about the Mean" (Amplitude Amplification).
3. Implement a 2-qubit search algorithm in Qiskit.

Important: The Speedup

The Quadratic Speedup

Classically, finding one specific item in an unsorted list of N items requires evaluating the list $N/2$ times on average, and N times in the worst case ($\mathcal{O}(N)$). Grover's Algorithm finds the item in roughly $\sqrt{N}$ steps ($\mathcal{O}(\sqrt{N})$). For a database of 1,000,000 items, a classical computer needs up to 1,000,000 checks. A quantum computer needs only about 1,000 checks!

2 Part A: The Anatomy of a Quantum Search

Grover's Algorithm always follows the same logical loop:

1. **Initialize:** Create an equal superposition of all possible answers.
2. **The Oracle:** Flip the phase (make it negative) *only* for the correct answer.
3. **The Diffuser:** Perform an "Inversion about the Mean" to shrink the wrong answers and amplify the correct one.
4. **Repeat:** Steps 2 and 3 are repeated about $\frac{\pi}{4}\sqrt{N}$ times (exactly once for $N = 4$).

Step-by-Step: The Math of Amplitude Amplification

The 2-Qubit Magic

We will search a database of $N = 4$ items (using 2 qubits). The winning item is $|11\rangle$. Let's trace the amplitudes.

1. **Superposition (H gates):**

$$|\psi_1\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

The average (mean) amplitude is $\frac{1}{2}$. Probabilities are 25% each.

2. **The Oracle (Marks $|11\rangle$):** The Oracle applies a negative phase to the winner.

$$|\psi_2\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

The new mean amplitude drops to $\frac{1}{4}$ (because three are +0.5 and one is -0.5).

3. **The Diffuser (Inversion about the Mean):** The Diffuser reflects every amplitude around the new mean ($\frac{1}{4}$). The formula is: New = $2 \times \text{Mean} - \text{Old}$.

- For wrong answers (+0.5): $2(\frac{1}{4}) - \frac{1}{2} = \frac{1}{2} - \frac{1}{2} = 0$.
- For the winner (-0.5): $2(\frac{1}{4}) - (-\frac{1}{2}) = \frac{1}{2} + \frac{1}{2} = 1$.

$$|\psi_3\rangle = 0|00\rangle + 0|01\rangle + 0|10\rangle + 1|11\rangle$$

Result: In exactly 1 step, the probability of measuring the winner $|11\rangle$ went from 25% to 100%!

3 Part B: Building Grover (IBM Quantum Composer)

Platform: IBM Quantum Composer

3.1 Exercise 1: Visualizing the Amplification

We will build the Oracle to mark $|11\rangle$. A gate that applies a phase only when both qubits are 1 is the **Controlled-Z (CZ)** gate.

Student Activity

Setup: A new Composer circuit has four qubits. Reduce it to two: in the OpenQASM editor change `qubit[4] q;` to `qubit[2] q;` and `bit[4] c;` to `bit[2] c;` (or use *Edit* $\rightarrow$ *Manage registers*). With four qubits $|11\rangle$ is shown as 0011.

Action:

1. Put q_0 and q_1 in superposition using H gates.
2. **The Oracle:** Apply a CZ gate connecting q_0 and q_1 . Look at the Q-Sphere: the state $|11\rangle$ is now Red (negative phase).
3. **The Diffuser:**
 - Apply H gates to both qubits.
 - Apply X gates to both qubits.
 - Apply a CZ gate between them.
 - Apply X gates to both qubits.
 - Apply H gates to both qubits.
4. **Look at the Probabilities bar chart.**

Expected Effect: The bar chart will show exactly 100% probability on the state $|11\rangle$. The algorithm successfully amplified the marked state and suppressed the others.

Note: these five Diffuser steps give the inversion about the mean of Part A with an overall minus sign (a global phase -1): the circuit ends in $-|11\rangle$ instead of $|11\rangle$. A global phase changes no probability, so the bar chart is the same.

4 Part C: Coding Grover's Algorithm (Qiskit)

4.1 Exercise 2: Python Implementation

Now we implement the 2-qubit Grover search in Qiskit to find the state $|11\rangle$.

Coding Task: Grover for $|11\rangle$

```
1 from qiskit import QuantumCircuit
2 from qiskit_aer import AerSimulator
3 from qiskit.visualization import plot_histogram
4
5 # 1. Setup
6 qc = QuantumCircuit(2, 2)
7
8 # 2. Initialization: Equal Superposition
9 qc.h([0, 1])
10 qc.barrier()
11
12 # 3. THE ORACLE (Marks state  $|11\rangle$ )
13 # A Controlled-Z gate flips the phase only if both qubits are 1
14 qc.cz(0, 1)
15 qc.barrier()
16
17 # 4. THE DIFFUSER (Amplitude Amplification)
18 # Step 4a: Apply H gates
19 qc.h([0, 1])
20 # Step 4b: Apply X gates (shifts the  $|00\rangle$  state to  $|11\rangle$ )
21 qc.x([0, 1])
22 # Step 4c: Apply CZ (marks the new  $|11\rangle$  state)
23 qc.cz(0, 1)
24 # Step 4d: Undo X and H
25 qc.x([0, 1])
26 qc.h([0, 1])
27 qc.barrier()
28
29 # 5. Measurement
30 qc.measure([0, 1], [0, 1])
31
32 # 6. Execution
33 simulator = AerSimulator()
34 job = simulator.run(qc, shots=1000)
35 counts = job.result().get_counts()
36 print(f"Search Results: {counts}")
```

Analysis: The output will be 100% ' 11 '. You found the hidden data in just 1 query!

5 Part D: Assessment (Quiz)

Q1. The Speedup

If you have an unsorted database of 10,000 items, roughly how many operations does Grover's Algorithm need to find the target?

- A) 10,000
- B) 5,000
- C) 100

Q2. The Division of Labor

In Grover's Algorithm, what is the specific job of the **Diffuser**?

- A) To "mark" the correct answer by flipping its phase to negative.
- B) To invert all amplitudes around the mean, which shrinks the wrong answers and boosts the marked correct answer.
- C) To put all qubits into an initial equal superposition.

Solutions: Q1: C ($\sqrt{10000} = 100$); Q2: B (The Oracle marks the answer, the Diffuser amplifies it).